

Controls Systems Integration Engineer with 6+ years of experience in automotive controls, system integration, and vehicle-level validation. Experienced in developing, integrating, and validating safety-critical automotive systems from concept to release

Education

• **Master of Science in Automotive Engineering** – Vehicle Dynamics & Controls
Clemson University - International Centre for Automotive Research

Aug '23 – Aug '25
Greenville, SC, USA

Professional Experience

Systems Test & Integration Engineer

Rivian Volkswagen Group Technologies – Vehicle Dynamics & Controls

Nov '25 – Present
Irvine, CA, USA

- Owned system-level and regression validation for vehicle motion and propulsion control software across R1, EDV, and R2 programs, supporting distributed and zonal E/E architectures from feature integration through production release.
- Translated vehicle dynamics control requirements into SIL, HIL/Dyno HIL, and in-vehicle verification activities, aligning test execution with program milestones and release-readiness criteria.
- Validated CAN/CAN FD, LIN, UDS, and Automotive Ethernet interfaces using DBC, LDF, ARXML, and PCAP data while automating Python-based test execution and post-processing workflows and maintaining traceability.
- Lead cross-functional root-cause investigations for vehicle-level software and ECU integration issues, drive and verify corrective fixes, and update automated/manual test plans to prevent recurrence in subsequent regression cycles.

Chassis Controls Software & Systems Intern

Ford Motor Company - Research & Advanced Engineering Division

May '25 – July '25
Dearborn, MI, USA

- Led deployment and real-time validation of a dual-redundant Steer-by-Wire platform from concept through integration, executing MIL, HIL, and proving-ground verification for motion-control functionality and robustness.
- Developed and integrated real-time Steer-by-Wire control software on dSPACE MicroAutoBox ECU, validating CAN interfaces, redundancy behavior, calibration changes, and vehicle response.
- Designed, built, and interpreted prototype wiring harnesses, schematics, and connector pinouts while supporting ECU bring-up and hardware/software debugging across bench, HIL, and vehicle environments.

Deep Orange 16 - Vehicle Dynamics & Controls Engineer / Graduate Research Assistant

Clemson University – International Centre for Automotive Research

Jan '24 – Apr '25
Greenville, SC, USA

- Developed physics-based vehicle dynamics control strategies including Direct Yaw Moment Control, supervisory brake-by-wire logic, and active suspension validation for a high-voltage 4-wheel hub-motor vehicle.
- Designed and validated MIL/HIL setups for brake-by-wire and active suspension systems, mapping ISO 26262-oriented fault responses to test cases, fault-injection scenarios, and fail-safe behavior.
- Led vehicle instrumentation, sensor integration, and dynamic testing on a BMW 535xi using oscilloscopes, DMMs, power supplies, and CAN/IMU/accelerometer/ride-height data to validate and refine system models.

Assistant Manager, Product Development – Chassis Testing & Validation

Royal Enfield (A unit of Eicher Motors Pvt. Ltd.)

Aug '18 – Jun '23
Chennai, Tamil Nadu, India

- Managed chassis systems integration and verification activities, resolving development issues through benchmarking, cross-functional collaboration, root-cause analysis, and validation of design changes.
- Defined and executed component-, system-, and vehicle-level validation plans from concept and prototype development through production release, including ABS calibration, performance validation, and regulatory compliance.

Skills

MATLAB/Simulink/Stateflow, Python, C/C++, Embedded C, CarSim, Linux/Bash/PowerShell; CAN/CAN FD, LIN, Ethernet, UDS, XCP; Vector CANalyzer/CANape, CANdb++, ASAP2 Studio, dSPACE, ATI Vision/VDA; MIL/SIL/HIL, ECU bring-up, test automation, OTA validation, CI/CD, Git/GitLab, Jenkins, SCons/Bazel fundamentals, JAMA/JIRA, ISO 26262.